

ENGS21 Group1 Analysis

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1 Theoretical Basis

In designing a wearable inhaler with a spring-driven (or similar mechanical) trigger for a liquid medication, several fundamental physics equations and relationships come into play. These include spring mechanics, fluid mechanics. The key equations are listed below.

1.1 Spring Mechanics

Hooke's Law:

$$F = k x \quad (1)$$

where F is the restoring force of the spring, k is the spring constant, and x is the displacement from equilibrium.

Pressure From Spring Force:

$$P = \frac{F}{A} \quad (2)$$

where P is the fluid pressure exerted by the spring force and A is the cross-sectional area over which the force is applied.

1.2 Fluid Mechanics

Orifice / Nozzle Flow Equation:

$$Q = C_d A \sqrt{\frac{2 \Delta P}{\rho}} \quad (3)$$

where Q is the volumetric flow rate, C_d is the discharge coefficient, A is the orifice area, ΔP is the pressure difference across the orifice, and ρ is fluid density.

2 Model and Assumptions

In our wearable inhaler concept, compressing a small spring provides the force needed to pressurize the liquid medication. We model this by Hooke's law, $F = k * x$, which tells us how much force the spring exerts for a given compression. That force is transmitted to the fluid through a piston or diaphragm of cross-sectional area A . The resulting fluid pressure is then $P = \frac{k * x}{A}$. Once the fluid reaches the requisite pressure P , it is forced through a small nozzle opening. We approximate the resulting flow rate with an orifice equation:

$$Q = C_d A \sqrt{\frac{2 \Delta P}{\rho}} = C_d A_{nozzle} \sqrt{\frac{2 (P - P_{ambient})}{\rho_{medicine}}} \quad (4)$$

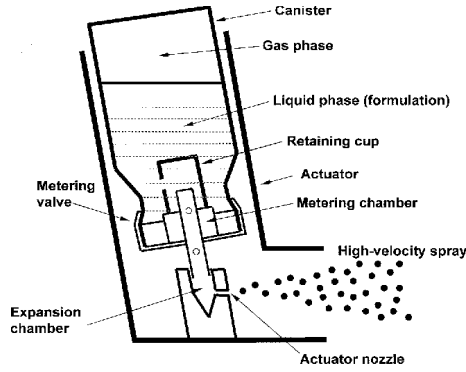


Fig. 1. Schematic of typical pressurized metered dose inhaler

Fig1. A sample sketch of inner mechanism of an inhaler

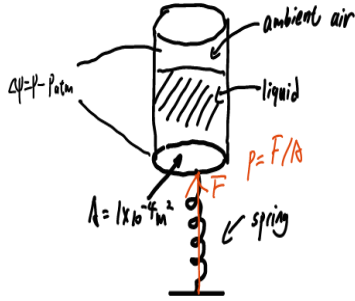


Fig2. Our model

2.1 Assumption

General Assumption:: Simplify the mechanical system to trigger the liquid spray as a single spring. And the energy loss is negligible (the pressure/force that user applies would be directly exerted on the fluid)

Incompressible Fluid: We assume the liquid medication behaves as an incompressible, Newtonian fluid. Any changes in density with pressure or temperature are negligible over the short actuation period.

Negligible Friction and Head Losses: We treat the flow passage and nozzle as having minimal frictional or viscous losses. While real designs have some pressure drop due to fluid friction and turbulence, our simplified approach keeps the model analytically manageable.

Uniform Spring Compression: We assume the spring is compressed uniformly and remains aligned.

Instantaneous Pressure Increase: Because actuation occurs rapidly, we take the pressure inside the fluid chamber to be nearly uniform at each instant. This neglects transient effects but is reasonable for a short-duration spray.

3 Analysis

3.1 Assumptions and Known values

Assume:

1. Spring constant: $k = 500 \text{ N/m}$
2. Compression distance: $x = 2 \text{ cm} = 0.02 \text{ m}$
3. Cross-sectional area of the fluid chamber/piston: $A = 1 \times 10^{-4} \text{ m}^2$. i.e 1cm x 1cm cross-sectional area
4. Ambient pressure: $P_{atm} = 1 \times 10^5 \text{ Pa}$
5. density of the liquid $\rho_{medicine} \approx 1000 \text{ kg/m}^3$ (close to water for a rough estimate).

3.2 Plug in numbers

3.2.1 Spring Force and Resulting Pressure

First, we calculate the force generated by the spring using Hooke's Law:

$$F = kx = (500 \text{ N/m}) \times (0.02 \text{ m}) = 10 \text{ N}.$$

Then, we find the resulting pressure on the fluid (gauge pressure) by dividing by the cross-sectional area:

$$P_{inhaler} = \frac{F}{A} = \frac{10 \text{ N}}{1 \times 10^{-4} \text{ m}^2} = 1 \times 10^5 \text{ Pa}.$$

Hence, the inhaler chamber pressure is $1 \times 10^5 \text{ Pa}$ **above** atmospheric pressure (about 1 bar gauge). In terms of absolute pressure, it is approximately

$$P_{absolute} \approx P_{atm} + P_{inhaler} = (1 \times 10^5 + 1 \times 10^5) \text{ Pa} = 2 \times 10^5 \text{ Pa}.$$

3.2.2 Potential Flow Rate through a Small Opening

If we assume the medication is expelled through a small opening, we can estimate the volumetric flow rate using a simplified nozzle-flow relationship:

$$Q = C_d A_{nozzle} \sqrt{\frac{2 \Delta P}{\rho_{medicine}}},$$

where C_d is the discharge coefficient (often around 0.6–0.8), A_{nozzle} is the nozzle cross-sectional area, and ΔP is the pressure difference (here, approximately $1 \times 10^5 \text{ Pa}$).

For instance, if $A_{nozzle} = 1 \times 10^{-7} \text{ m}^2$ (about 0.3 mm diameter) and $C_d = 0.70$, then:

$$Q = 0.70 \times 1 \times 10^{-7} \text{ m}^2 \times \sqrt{\frac{2 \times (1 \times 10^5 \text{ Pa})}{1000 \text{ kg/m}^3}}.$$

Inside the square root,

$$\frac{2 \times 10^5}{1000} = 200 \quad \longrightarrow \quad \sqrt{200} \approx 14.14.$$

Hence,

$$Q \approx 0.70 \times 1 \times 10^{-7} \text{ m}^3/\text{s} \times 14.14 = 9.9 \times 10^{-7} \text{ m}^3/\text{s}.$$

Converting to milliliters per second:

$$1 \text{ m}^3 = 1,000,000 \text{ ml} \implies 1 \times 10^{-6} \text{ m}^3 = 1 \text{ ml}.$$

Thus,

$$Q \approx 0.99 \text{ ml/s}.$$

4 Usefulness

- The required spring force to achieve a gauge pressure of $1 \times 10^5 \text{ Pa}$ (about 1 bar above atmospheric) is **10 N**.
- At this pressure, a 0.3 mm-diameter opening can deliver about **1 ml of fluid per second**.

Initially, approaching the task of designing an inhaler with an engineered mechanism to pump albuterol seemed daunting, but now that we have broken down the problem into a system of equations with tangible metrics we are confident we can optimize this product for the user.

Furthermore, this analysis helped us visualize the relationship between the force applied by the user and the flow rate of the medicine, and this relationship can be adjusted by the diameter of the nozzle. Based on this realization, we've decided to adjust our inhaler design to include an adjustable mechanism for the diameter of the mouthpiece to account for variability in force applied by the user.